

BeatWeaver: A Real-Time Music-Conditioned Framework for Adaptive Humanoid Control

Chengjun Ouyang
University College London
London, United Kingdom

Abstract—Humanoid animation conveys expressive intent through coordinated whole-body movement and its timing. In interactive settings, this movement must adapt to changing external cues while remaining coherent over time and consistent with the constraints of the animated body. To address these demands in music-driven performance, we present BeatWeaver, a causal retrieval-and-synthesis system for the 29-joint Unitree G1 representation. Streaming audio is compared with a paired music-motion catalogue using learned music embeddings and explicit rhythm, tempo, and activity descriptors. Offline, human motions are retargeted to the humanoid using General Motion Retargeting (GMR). Our collision-aware smoothing procedure then refines these trajectories with a receding-horizon quintic planner that preserves accepted position, velocity, and acceleration. Online, a transition planner searches compatible exit-entry frame pairs within a musically relevant shortlist and constructs joint-state bridges subject to continuous polynomial bounds. Boundary-aware time scaling and modulation connect authored trajectories to these bridges without predicting future audio. A shared Hermite representation links offline validation, online planning, and playback. The system produces inspectable kinematic motion references.

Index Terms—music-driven motion, humanoid animation, motion retrieval, motion retargeting, trajectory smoothing, transition planning

I. INTRODUCTION

Musical rhythm provides timing cues for movement, while timbre, style, and activity influence its expressive character. These complementary signals make music an appealing input for interactive humanoid performance. A useful system must respond to observed audio while maintaining coherent movement as the music and selected choreography change. For a humanoid representation, this also requires attention to joint ranges, derivative limits, and geometry.

Music-conditioned dance generation has advanced through paired datasets and transformer models, learned choreographic codes, and diffusion-based synthesis [1]–[3]. Recent work addresses causal streaming and direct audio-conditioned humanoid motion [4], [5]. These approaches motivate increasingly responsive interfaces, but generation is not the only useful design choice. When a finite collection of choreographies is available, retrieval permits offline inspection and preparation of each motion, explicit selection rules, and traceable relationships between source data and output.

Retrieval alone leaves two continuity problems unresolved. Within a clip, retargeting and collision correction can introduce abrupt stops or derivative changes. Across clips, similar poses can still have incompatible velocities and accelerations.

A blend that interpolates positions does not automatically preserve the states of the trajectories it joins. Moreover, a trajectory validated with one interpolation rule may violate constraints when a different rule is used for playback. These issues motivate treating motion as a sequence of joint states, rather than a sequence of poses.

We present *BeatWeaver*, a music-conditioned retrieval and kinematic synthesis system that addresses both continuity problems through a shared trajectory representation. Music retrieval identifies a relevant motion neighbourhood. Offline state-aware smoothing prepares GMR-retargeted clips. On-line state-to-state planning connects the resulting trajectories through joint position, velocity, and acceleration at their boundaries. Paired exit-entry search also allows a short terminal portion of a source clip to be trimmed when an earlier boundary is more compatible.

The paper describes three contributions:

- A causal music-to-motion pipeline combining semantic retrieval with explicit temporal and activity compatibility.
- An offline, collision-aware smoothing procedure that carries joint position, velocity, and acceleration through replanning and stores the same states used by playback.
- An online transition method combining nonlinear exit-entry scoring, checked Hermite bridges, and boundary-aware music adaptation.

II. RELATED WORK

A. Music-Conditioned Motion

AI Choreographer introduces AIST++ and FACT, pairing 3D dance with music and learning cross-modal temporal relationships [1]. Bailando represents dance using a learned choreographic memory and composes motion units with an actor-critic transformer [2]. EDGE uses diffusion for editable music-conditioned dance and incorporates contact-related physical plausibility considerations [3]. These methods generate human motion; a humanoid application additionally needs an embodiment mapping and an execution strategy.

DiscoForcing explicitly addresses causal, bounded-latency audio-driven character control [4]. RoboPerform investigates expressive humanoid motion driven by music and speech [5]. BeatWeaver occupies a complementary point in this design space: it selects and connects existing robot-space references instead of training an online choreography generator. Its expressive coverage is consequently limited by the catalogue, while its selection and transition constraints remain explicit.

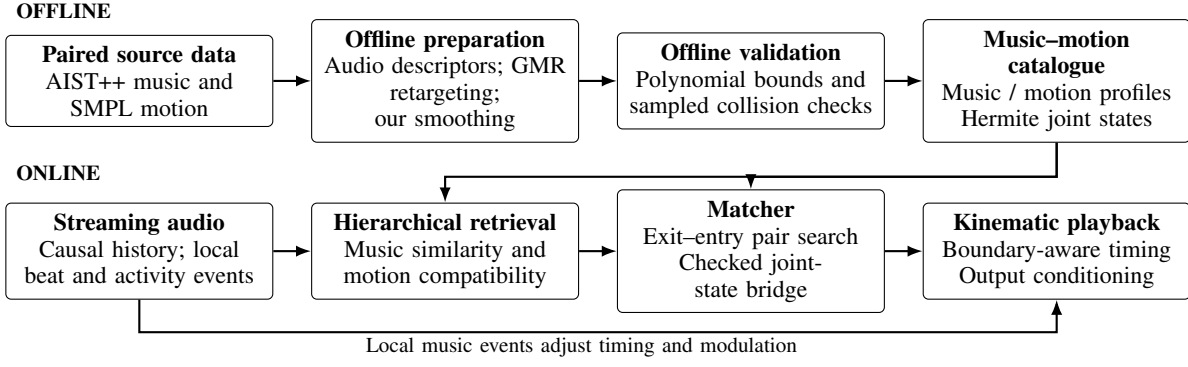


Fig. 1. Overview of the BeatWeaver pipeline

B. Music Representation and Motion Retrieval

Music embeddings learned from Discogs metadata capture relationships that complement engineered acoustic descriptors [6]. Signal-processing libraries such as librosa provide tempo, onset, and spectral features [7]. BeatWeaver combines these feature families because stylistic similarity does not by itself imply a usable motion speed or matching event density.

Motion graphs assemble captured motion through transitions between compatible states [8]. Learned motion matching develops a compact learned alternative to database-based motion matching [9]. BeatWeaver shares the use of reusable motion data, but separates musical relevance from instantaneous boundary compatibility. It retrieves a music-conditioned short-list and plans joint-state bridges online rather than constructing a fixed all-pairs motion graph.

C. Retargeting and Constrained Trajectories

SMPL supplies a common human-motion representation [10], while General Motion Retargeting (GMR) maps human motion to humanoid embodiments. The GMR study shows that reference-motion quality affects downstream tracking [11]. BeatWeaver uses GMR as its retargeter and applies our collision-aware, state-preserving smoothing procedure to its output. The resulting trajectories store joint position, velocity, and acceleration for playback and transition planning. This postprocessing is distinct from the published GMR method.

Position, velocity, and acceleration are also central to constrained trajectory generation. Ruckig constructs synchronized trajectories under derivative constraints for arbitrary target states [12]. BeatWeaver provides an optional Ruckig backend. Its default method uses quintic Hermite curves with an explicit duration search. The focus here is the integration of state-preserving clip preparation, musical retrieval, and online boundary selection, rather than a new general-purpose trajectory solver.

III. METHOD

A. Problem Formulation and System Overview

Let $\mathcal{A}_{\leq t}$ denote audio observed by wall time t . The catalogue contains music descriptors, motion profiles, and robot

trajectories:

$$\mathcal{D} = \{(\mathbf{x}_i, \mathbf{m}_i, M_i)\}_{i=1}^N. \quad (1)$$

Here N is the number of catalogue entries. Each trajectory has a floating-base pose and joint states $\mathbf{s}_{i,k} = (\mathbf{q}_{i,k}, \mathbf{v}_{i,k}, \mathbf{a}_{i,k})$, where $\mathbf{q}_{i,k} \in \mathbb{R}^{29}$ and derivatives are defined in authored time. The system chooses a motion index, its playback time, and, when needed, a transition to another motion using only $\mathcal{A}_{\leq t}$ and stored state.

Fig. 1 separates offline preparation from online execution. Offline, paired AIST++ music and SMPL motion are converted into audio descriptors and G1 reference trajectories through GMR retargeting followed by our smoothing procedure. Online, audio retrieval, candidate loading, and transition search run outside the foreground playback path. The foreground consumes completed plans and emits kinematic poses to MuJoCo [13]. The default matcher update rate is 60 Hz. This is a configuration, not a hard real-time guarantee.

B. Music Descriptors and Hierarchical Retrieval

Catalogue audio is converted to mono at 16 kHz and analysed in six-second windows with a two-second hop. Median removal and percentile amplitude normalization reduce sensitivity to recording gain. The learned backend produces a 1,280-dimensional Discogs EffNet embedding $\mathbf{e}$ and 400 style activations $\mathbf{g}$. A 24-dimensional rhythm-timbre vector $\mathbf{r}$ combines tempo, onset and beat statistics, spectral flux, percussive and band energy ratios, MFCCs, and spectral contrast.

The matcher maintains a causal history of up to 30 s, beginning analysis after 2 s. It requests retrieval at one-second intervals when the worker is free. Unlike the six-second catalogue segmentation, the query descriptor summarizes the currently available history; learned patch outputs are averaged. Thus longer history offers more context but can retain evidence from earlier musical passages. No future waveform is used, and busy workers do not receive an accumulating queue of queries.

Embedding and rhythm vectors are standardized using fixed catalogue statistics and normalized to unit length. Style vectors are unit-normalized without standardization. For normalized

vectors, define $s_f(q, n) = (1 + \bar{\mathbf{f}}_q^\top \bar{\mathbf{f}}_n)/2$. Segment similarity is

$$S_n = 0.70s_e(q, n) + 0.20s_r(q, n) + 0.10s_g(q, n). \quad (2)$$

If style vectors are unavailable, the remaining weights are renormalized. Track evidence is the mean of its three highest segment scores, or all scores when fewer than three exist. The default style-first policy combines track-based genre evidence with broad style-label priors and restricts retrieval to the leading genre, admitting the runner-up when their scores are close. Weak beat evidence or persistent non-dance style evidence causes abstention from new music-driven selection.

Motion profiles add source tempo b_i , velocity-derived activity, and the density and regularity of salient poses. These poses are detected as prominent valleys in whole-body source-motion velocity. For query tempo b_q , consider $\{b_q/b_i, b_q/(2b_i), 2b_q/b_i\}$ to accommodate half/double-time ambiguity. Choose the ratio r_i closest to one in log space within the configured speed interval, default $[0.55, 1.60]$. If no ratio is admissible, exclude the motion. Tempo compatibility is

$$S_{\text{tempo},i} = \exp\left[-\frac{1}{2}(\log r_i/0.28)^2\right]. \quad (3)$$

Key-pose compatibility compares playback-scaled key-pose density with a weighted combination of beat and onset rates, together with an interval regularity term. Activity compatibility is one minus the absolute difference between normalized audio and motion activity. The final score is

$$S_{\text{motion},i} = 0.50S_{\text{tempo},i} + 0.35S_{\text{key},i} + 0.15S_{\text{activity},i}, \quad (4)$$

$$S_i = 0.65S_{\text{track},i} + 0.35S_{\text{motion},i}. \quad (5)$$

C. State-Preserving Offline Motion Smoothing

GMR transfers each SMPL sequence to the target humanoid. The input to our smoothing procedure is the GMR-retargeted trajectory before it has undergone our smoothing. We first construct a collision-aware positional guide from this trajectory using joint displacement optimization, range and speed bounds, and linearized distance constraints. Nonlinear geometry checks restrict unsafe steps.

The guide is not the final playback trajectory. Starting from an accepted state s_0 , a receding-horizon planner fits a quintic joint curve over up to 12 source frames. For horizon duration T and normalized time $u \in [0, 1]$, write

$$Q(u) = \sum_{k=0}^5 \mathbf{c}_k u^k, \quad \mathbf{z} = [\mathbf{q}_T^\top, T\mathbf{v}_T^\top, T^2\mathbf{a}_T^\top]^\top. \quad (6)$$

The initial position, velocity, and acceleration are fixed; coefficients are affine in the optimized terminal state $\mathbf{z}$. Future guide states are tracking objectives rather than hard endpoint constraints. With $Q^{(d)}$ denoting the d th derivative with respect to u , the implemented objective has the form

$$\begin{aligned} \min_{\mathbf{z}} \quad & \sum_{\ell} \sum_{d=0}^2 \omega_d \left\| Q^{(d)}(u_\ell) - T^d \hat{\mathbf{s}}_\ell^{(d)} \right\|^2 \\ & + \frac{\lambda_j}{5} \sum_{r=0}^4 \left\| T^{-3} Q^{(3)}(r/4) \right\|^2, \end{aligned} \quad (7)$$

where $\hat{\mathbf{s}}_\ell^{(0,1,2)}$ denotes guide position, velocity, and acceleration, $\omega = (1, 0.1, 0.005)$, and λ_j sets a soft physical-time jerk cost. The jerk term is sampled regularization, not a hard jerk bound.

Bernstein control-point bounds constrain position, speed, and acceleration throughout the curve. Bounds are applied on four equal subintervals to reduce conservatism. Signed-distance constraints are relinearized during sequential quadratic optimization, and a short terminal continuation reserves room for motion beyond the planning horizon. Candidate curves undergo nonlinear collision sampling and continuous joint-extrema checks before acceptance.

Only one source-frame interval is committed before replanning. Its ending position, velocity, and acceleration become the next fixed initial state. Shorter horizons are tried when necessary. If replanning fails, the remaining previously validated curve may be consumed; an instantaneous zero-velocity hold is not inserted into the generated trajectory. When no validated continuation exists, generation fails. At clip initialization only, estimated derivatives may be reduced if no incoming state was supplied, and the adjustment is recorded.

The smoothed trajectories retain the GMR input's frame count and timing and are stored as float64 position, velocity, and acceleration arrays. A checksum binds these arrays to the frame rate. Playback and transition planning reuse the saved states after joint-name mapping rather than re-estimating derivatives from poses. The serialized piecewise Hermite curves are checked again for joint bounds and dense sampled clearance in both configured robot models. This aligns validation with the interpolation actually used at authored speed. It does not establish continuous collision avoidance or validate later time warping.

D. Musical Shortlists and Exit-Entry Search

The matcher retrieves up to 20 motions and retains at most ten alternatives whose final and music scores lie within 0.05 and 0.08 of the leading motion, respectively. The active motion is excluded from this normal shortlist. The shortlist is frozen for preparation, and a successful plan is retained. This policy avoids replacing an imminent transition with every new ranking.

For each shortlisted motion j , eligible entries include every authored frame with phase $k/(N_j - 1) \leq 0.2$. Source exits include reachable frames in the final two authored seconds of the current clip. Invalid joint boundary states are filtered first. An early exit must still allow the boundary fade to begin before that exit and before the original terminal fade. Retaining the original phase denominator while storing a separate stop time prevents trimming from rescaling the source motion.

Each exit-entry pair is compared using joint pose, joint velocity, binary foot-contact state, and root features. For a nonnegative discrepancy d with soft tolerance τ , define

$$P(d; \tau) = r^2 + \kappa \max(r - 1, 0)^2, \quad r = d/\tau. \quad (8)$$

We use $\kappa = 4$. This penalty is sensitive below the tolerance and grows more strongly above it; the tolerance is not a hard

rejection threshold. Pose and velocity penalties are averaged across joints, using tolerances of 5% of joint range and 10% of joint speed limit. The contact term uses the number of mismatched feet with tolerance one. Root height, tilt, linear velocity, and angular speed discrepancies use tolerances of 0.025 m, 0.075 rad, 0.25 m/s, and 0.5 rad/s, respectively. Their penalties are averaged to form C_r . The total cost is

$$C_{\text{pair}} = \frac{0.45C_q + 0.20C_v + 0.20C_c + 0.10C_r}{0.95}. \quad (9)$$

Acceleration enters bridge feasibility, not this ranking cost. Music determines the shortlist and resolves ties; it is not an additional geometric penalty. Further ties prefer later exits, earlier entries, and a deterministic motion identifier order. Contact agreement is a heuristic, not a contact-force constraint.

Candidate rows are vectorized per exit and merged in sorted order, avoiding a full exit-by-entry-by-joint tensor. Bridge construction is attempted in this order until a feasible pair is found. The resulting choice is the first successful ranked pair encountered within the preparation deadline, rather than a globally optimal trajectory.

E. Checked State-to-State Bridges

Given an exit state (q_0, v_0, a_0) and an entry state (q_1, v_1, a_1) , a quintic Hermite bridge of duration T satisfies all six endpoint conditions. With derivatives specified in physical time, its normalized-time coefficients are

$$\begin{aligned} c_0 &= q_0, & c_1 &= T v_0, & c_2 &= \frac{1}{2} T^2 a_0, \\ d &= q_1 - c_0 - c_1 - c_2, \\ h &= T v_1 - c_1 - 2c_2, \\ g &= T^2 a_1 - 2c_2, \\ c_3 &= 10d - 4h + \frac{1}{2}g, \\ c_4 &= -15d + 7h - g, \\ c_5 &= 6d - 3h + \frac{1}{2}g. \end{aligned} \quad (10)$$

For each joint, the planner checks

$$q_{\min} \leq q(t) \leq q_{\max}, \quad |\dot{q}(t)| \leq v_{\max}, \quad |\ddot{q}(t)| \leq a_{\max} \quad (11)$$

throughout $[0, T]$. A jerk bound is also checked when explicitly configured; the default Hermite configuration has no finite jerk limit.

Duration search starts with a displacement-based estimate and coarse geometric steps. If those fail, a refined grid with spacing no greater than 2% covers the interval from the minimum duration, default 0.35 s, to 10 s. Nonzero endpoint derivatives make feasibility nonmonotonic in duration, so the displacement estimate is not treated as a lower bound. Failure of this finite search does not prove that every possible duration is infeasible.

Polynomial checks use sampled rejection followed by Bernstein convex-hull bounds and bounded subdivision. Uncertain cases fall back to evaluating endpoints and real stationary points in the interval. Sampled tests can reject a curve but cannot alone authorize it. The optional Ruckig backend instead

constructs a synchronized jerk-limited trajectory and checks its position extrema; it requires finite jerk limits for every joint.

Preparation checks cancellation and a conservative remaining-playback deadline between score rows, bridge attempts, and duration trials. Native solver calls are not preemptible. If no shortlisted transition is found, a checked replay to frame zero of the current motion is attempted. One expanded-shortlist retry may then use new candidates and wider musical margins while time remains. If no plan is ready at the terminal state, playback holds that pose and records the failure. This terminal hold lies outside the smooth-transition contract; its onset need not preserve a nonzero terminal velocity.

F. Boundary-Aware Music Adaptation and Playback

An authored joint curve is $Q(s)$, where s is clip time. Music-dependent time scaling changes its physical derivatives:

$$\dot{q} = Q'(s)\dot{s}, \quad \ddot{q} = Q''(s)\dot{s}^2 + Q'(s)\ddot{s}. \quad (12)$$

Thus a bridge planned from authored derivatives cannot simply meet a clip at an arbitrary music-dependent rate. BeatWeaver fades the rate toward authored speed and fades pose modulation toward zero near both boundaries.

Let s_a and s_b be the entry and selected exit times and h the boundary window, default 0.5 s and reduced for short intervals. Define

$$H(u) = 10u^3 - 15u^4 + 6u^5, \quad u \in [0, 1], \quad (13)$$

$$w(s) = \min \left\{ H \left(\text{clip}_{[0,1]} \frac{s-s_a}{h} \right), H \left(\text{clip}_{[0,1]} \frac{s_b-s}{h} \right) \right\}. \quad (14)$$

Playback integrates

$$\begin{aligned} \dot{s} &= 1 + w(s)(r_{\text{music}} - 1), \\ q_{\text{target}} &= Q(s) + w(s)\Delta q_{\text{music}}. \end{aligned} \quad (15)$$

The beat controller updates the bounded music rate and timing error without teleporting the sampler to a new phase. Near boundaries, accumulated phase correction is cleared. Because the window and its first two derivatives vanish at each join, playback approaches the authored state under regular bounded music modulation. The bridge then advances in wall time while the target clip's entry time remains fixed. On completion, the target resumes from that entry with wall-time overshoot carried forward.

This construction gives matching joint position, velocity, and acceleration at successful ideal joins before downstream conditioning. Root position and orientation are aligned and blended separately; they do not inherit the joint derivative guarantee. Stateful output limits and optional runtime collision handling can further modify emitted poses, so final-output continuity and constraint satisfaction require their own measurements.

G. Implementation Scope

The prototype uses named-joint G1 trajectories, NumPy-based scoring, ONNX audio inference, and MuJoCo kinematic playback. Retrieval and motion loading support separate

worker processes; pair search uses a background scoring worker. Generation identifiers reject stale plans, and early-exit plans arriving after their fade deadline are discarded. Asynchronous execution avoids deliberately waiting for planning in the playback loop, but does not establish a worst-case execution-time bound.

The online matcher loads trajectories processed by our offline smoothing procedure, together with their saved joint states. Motion provenance and saved-state checks verify consistency with the validated trajectory representation. Offline joint-curve bounds concern authored speed. Sampled self-collision checks do not certify inter-clip bridges, root dynamics, balance, ground-contact forces, or hardware tracking.

IV. EXPERIMENTS

TODO

V. RESULTS AND DISCUSSION

TODO

VI. CONCLUSION

BeatWeaver combines causal music retrieval with state-aware humanoid motion synthesis. Its central design choice is to use a consistent joint-state representation across offline smoothing, transition planning, and playback. Our offline smoothing procedure refines GMR-retargeted motions while carrying accepted derivatives through collision-aware replanning. The matcher searches musically relevant exit–entry pairs and joins them with checked trajectories. Boundary-aware adaptation reconciles music-dependent playback with authored bridge states. Evaluation of musical relevance, delivered smoothness, transition availability, and runtime performance remains to be completed. Physical deployment further requires a dynamically validated tracking controller.

REFERENCES

- [1] R. Li, S. Yang, D. A. Ross, and A. Kanazawa, “Ai choreographer: Music conditioned 3d dance generation with aist++,” in *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*. Ieee, 2021, pp. 13 381–13 392.
- [2] L. Siyao, W. Yu, T. Gu, C. Lin, Q. Wang, C. Qian, C. C. Loy, and Z. Liu, “Bailando: 3d dance generation by actor-critic gpt with choreographic memory,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 11 050–11 059.
- [3] J. Tseng, R. Castellon, and K. Liu, “Edge: Editable dance generation from music,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 448–458.
- [4] K. Ji, B. Qian, B. Wu, K. Chen, Y. Shi, and J. Wang, “Discoforcing: A unified framework for real-time audio-driven character control with diffusion forcing,” *arXiv preprint arXiv:2605.28491*, 2026.
- [5] Z. Li, C. Chi, Y. Wei, B. Zhu, T. Huang, Z. Sun, Y. Peng, P. Wang, Z. Wang, F. Liu *et al.*, “Do you have freestyle? expressive humanoid locomotion via audio control,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 956–965.
- [6] P. Alonso-Jiménez, X. Serra, and D. Bogdanov, “Music representation learning based on editorial metadata from discogs,” in *Rao P, Murthy H, Srinivasamurthy A, Bittner R, Caro Repetto R, Goto M, Serra X, Miron M, editors. Proceedings of the 23rd International Society for Music Information Retrieval Conference (ISMIR 2022); 2022 Dec 4-8; Bengaluru, India.[Canada]: International Society for Music Information Retrieval; 2022. p. 825-33. International Society for Music Information Retrieval (ISMIR)*, 2022.
- [7] B. McFee, C. Raffel, D. Liang, D. P. Ellis, M. McVicar, E. Battenberg, O. Nieto *et al.*, “librosa: Audio and music signal analysis in python.” *SciPy*, vol. 2015, no. 18-24, p. 7, 2015.
- [8] L. Kovar, M. Gleicher, and F. Pighin, “Motion graphs,” in *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*, 2023, pp. 723–732.
- [9] D. Holden, O. Kanoun, M. Perepichka, and T. Popa, “Learned motion matching,” *ACM Transactions on Graphics (ToG)*, vol. 39, no. 4, pp. 53–1, 2020.
- [10] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black, “Smpl: A skinned multi-person linear model,” in *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*, 2023, pp. 851–866.
- [11] J. P. Araujo, Y. Ze, P. Xu, J. Wu, and C. K. Liu, “Retargeting matters: General motion retargeting for humanoid motion tracking,” *arXiv preprint arXiv:2510.02252*, 2025.
- [12] L. Berscheid and T. Kröger, “Jerk-limited real-time trajectory generation with arbitrary target states,” in *Proc. Robotics: Science and Systems*, 2021.
- [13] E. Todorov, T. Erez, and Y. Tassa, “Mujoco: A physics engine for model-based control,” in *2012 IEEE/RSJ international conference on intelligent robots and systems*. IEEE, 2012, pp. 5026–5033.